

path. It uses environment variables that are mapped to `java.class.path`. It is also implemented in Java by the `sun.misc.Launcher$AppClassLoader` class.

```

• // Java code to demonstrate Class Loader subsystem
• public class Test {
•     public static void main(String[] args)
•     {
•         // String class is loaded by bootstrap
loader, and
•         // bootstrap loader is not Java object,
hence null
•         System.out.println(String.class.getClass-
Loader());
•
•         // Test class is loaded by Application
loader
•         System.out.println(Test.class.getClass-
Loader());
•     }
• }

```

The JVM uses the Delegation-Hierarchy principle to load classes. The System class loader delegates load requests to the Extension class loader, and the Extension class loader delegates requests to the Bootstrap class loader. If a class is found in the bootstrap path, it is loaded; otherwise, the request is again transferred to the Extension class loader and then to the System class loader. If the System class loader fails to load the class, then a runtime exception of `java.lang.ClassNotFoundException` is returned.

JVM Memory can be divided into four major parts:

1. **Method Area:** This area stores all class-level information such as the class name, immediate parent class name, method and variable information, and static variables. Prior to Java 8, static variables were stored in the method area, but now they are stored in the Heap area. There is only one method area per JVM, and it is a shared resource.
2. **Heap Area:** The heap area stores information for all objects in the program. There is also one Heap Area per JVM, and it is a shared resource.
3. **Stack Area:** The stack area stores information related to the run-time stack of each thread. Each block of this stack is known as an activation record or stack frame, which stores method calls. All local variables of that method are stored in their corresponding frame. After a thread terminates, its run-time stack will be destroyed by the JVM. It is not a shared resource.